

Project 13: Multi-Indicator Trading Signal System

Objective: Build a comprehensive technical analysis system that calculates multiple indicators, generates trading signals, and evaluates their effectiveness across different market conditions.

Requirements:

1. Select at least 4 different stocks from various sectors and download 2 years of historical data.
2. Implement the following technical indicators for each stock:
 - Simple Moving Average (SMA) with two different periods (e.g., 20-day and 50-day)
 - Exponential Moving Average (EMA) with two different periods
 - Relative Strength Index (RSI)
 - Bollinger Bands
 - MACD (Moving Average Convergence Divergence)
 - Stochastic Oscillator
3. Create functions to generate trading signals based on:
 - Moving average crossovers (SMA and EMA)
 - RSI overbought/oversold conditions
 - Bollinger Band breakouts
 - MACD line and signal line crossovers
 - Stochastic oscillator crossovers
4. Develop a signal combination framework that:
 - Tracks when multiple indicators generate the same signal